

Personal project: family expense analysis & visualisation

Objective

End-to-end personal project covering the full data workflow (collection, modelling, cleaning and visualisation) using personal data as a reliable and immediately verifiable source.

Data sources

Shopping tickets (PDF & paper) and bank data (Excel) processed in Power Query in both Excel and Power BI.

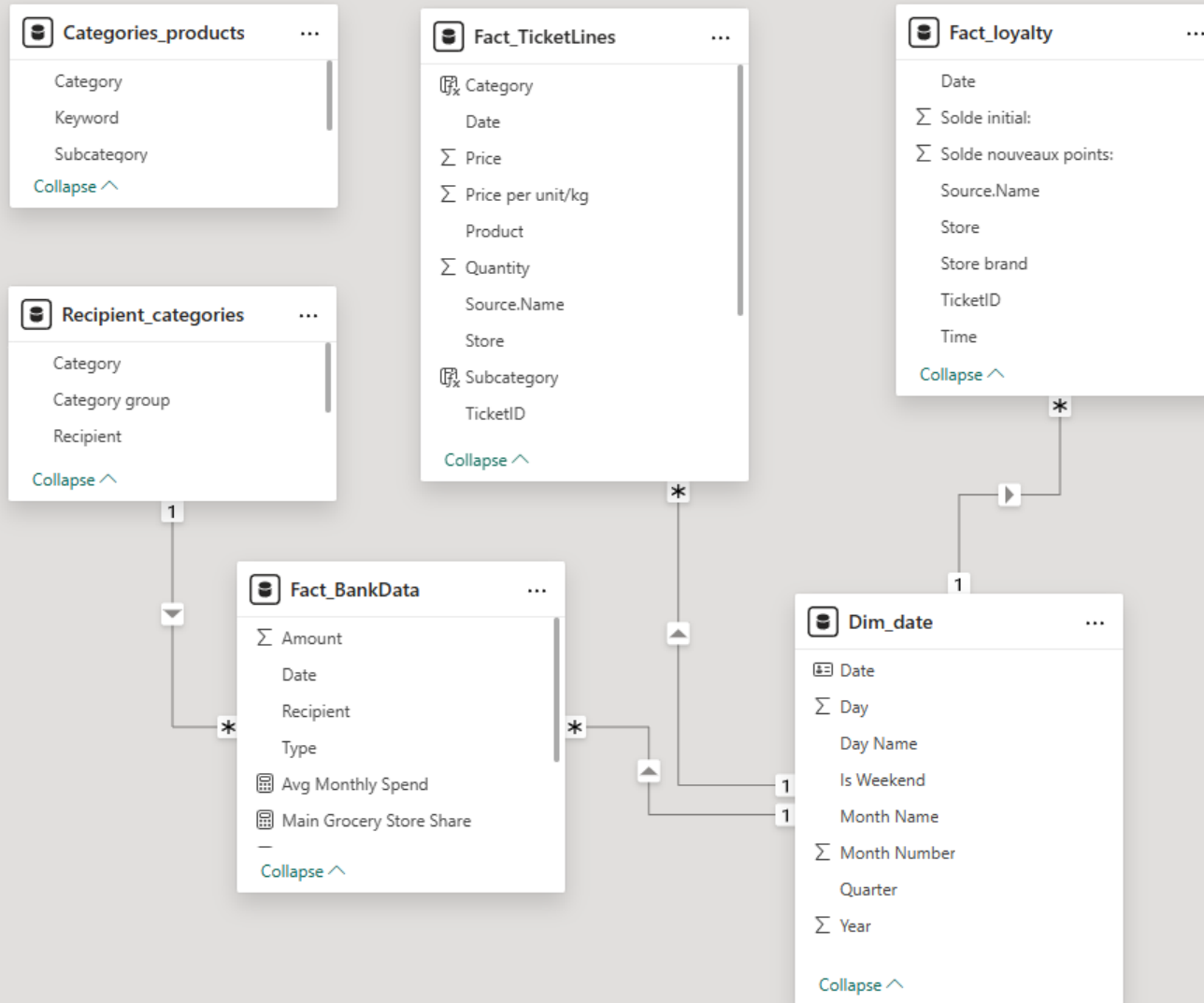
Data model

Multi-source data model with multiple fact tables connected through a shared date dimension. Two sources were intentionally kept separate to reflect different levels of granularity: bank data for overall spending patterns and ticket data for product-level analysis. Reporting covers January to April 2026.

Key insights:

- 20% of household spending is concentrated at one retailer
- Fish has the highest cost per unit among regularly purchased items
- Shopping behaviour varies significantly by day of the week

Personal Project: data model



Note: 'Categories_products' are assigned by keyword matching and not by a direct table relationship.

Personal Project: household spending overview

Total spending



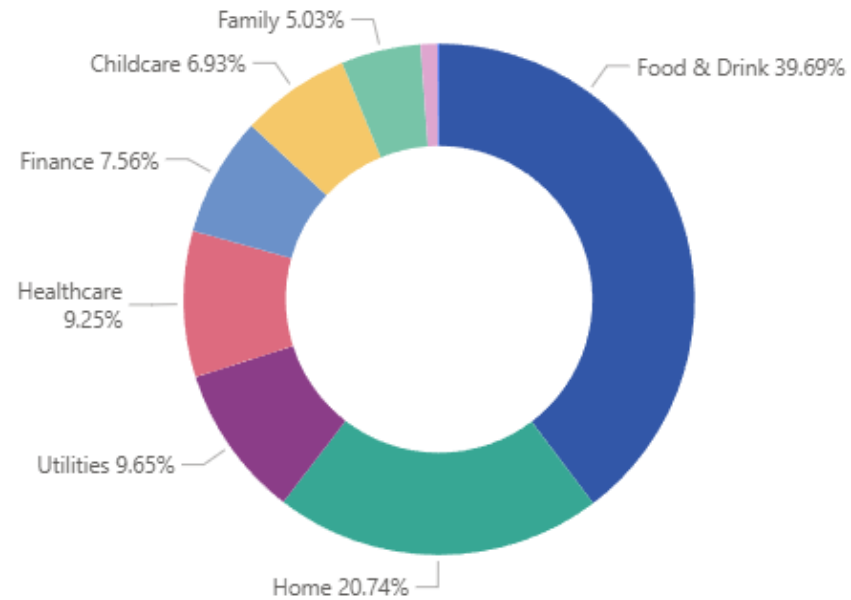
Top spending category

Food & Drink

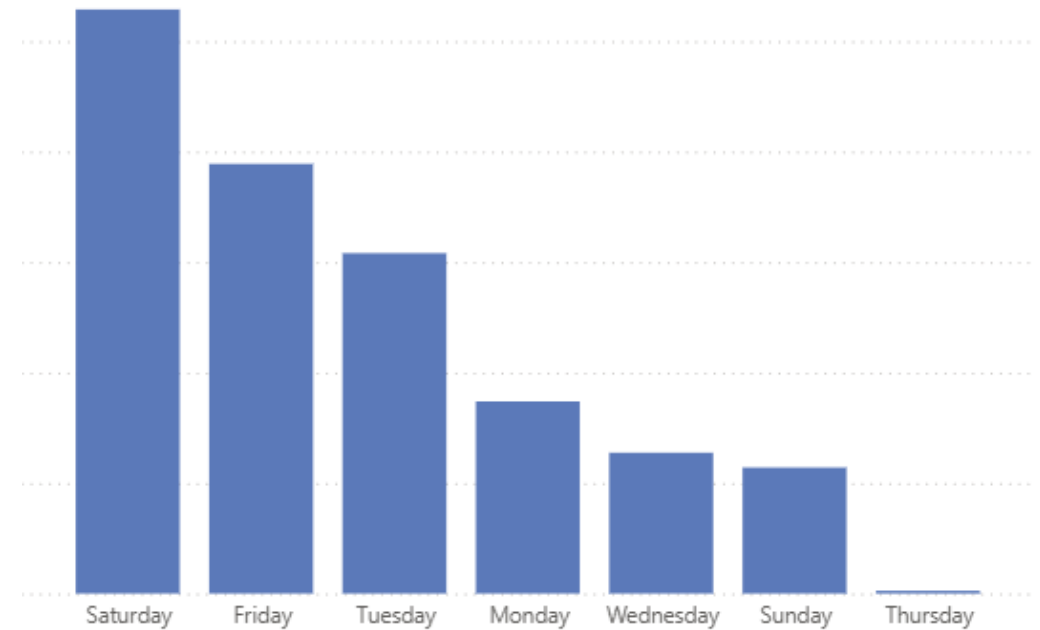
Main grocery store share

19.46%

Where our money goes



Busiest shopping days



Personal Project: shopping behaviour & loyalty points

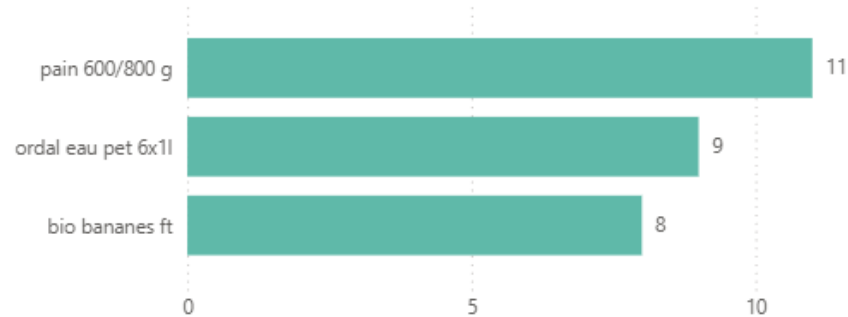
Top spending category

Food & Drink

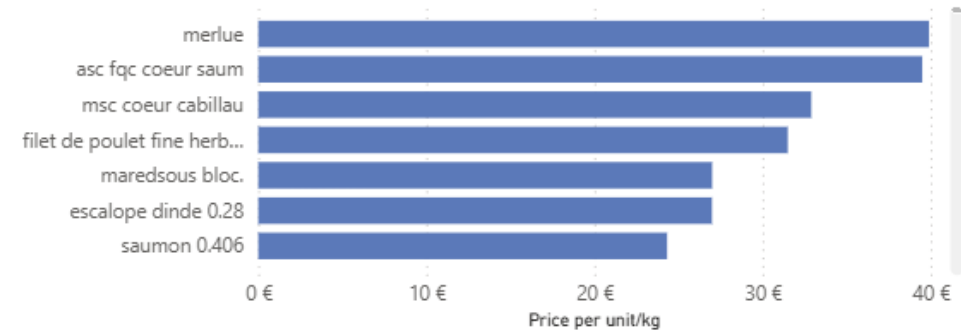
Main grocery store share

19.46%

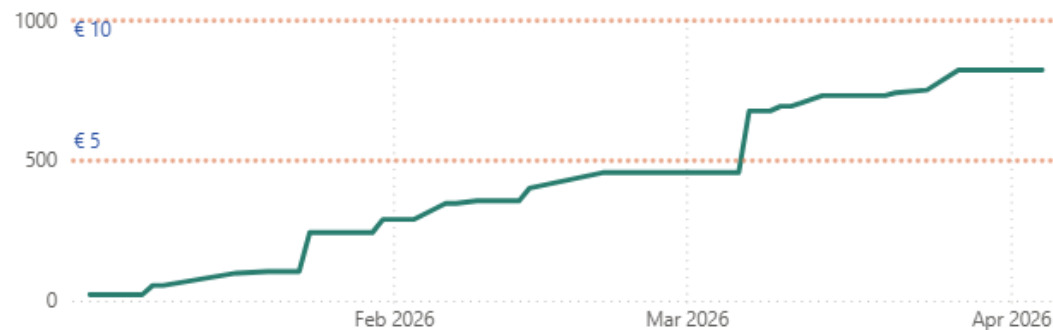
Most frequently bought items



Fish weighs a lot on our bill



Main grocery store: points adding up



Redeemable now

5 €

Points to next reward

177